Improving Image Retrieval with Semantic Classification Using Relevance Feedback

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Abstract

In this paper, we investigate the combination of image semantic classification with image retrieval. A flexible scheme is proposed to take advantage of image classification information, which may be obtained manually or automatically, to enhance image retrieval. In this scheme, a semantic feature vector is composed for an image based on its class membership information, and is combined with low-level features in image retrieval. Relevance feedback techniques are also used to adjust both the semantic feature and low-level features of the query in order to better reflect the user's intention. Experimental results on a collection of 10,000 images with manual classification demonstrate the effectiveness of the proposed method.

Keywords

Image retrieval, image classification, relevance feedback, semantic feature, and class membership

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1. Introduction

The performance of most content-based image retrieval (CBIR) systems [6,9,12] is limited due to the gap between semantic-level concepts and low-level features, and the subjectivity of human visual perception. One approach to improving this situation is to employ relevance feedback (RF) techniques [4,10,11]. In the relevance feedback process, the system dynamically learns and refines the query and retrieval results to capture the user's intention. However, most of RF methods in CBIR systems are not as effective as in text information retrieval systems, due to the fact that refining low-level features to map semantic content is often difficult. Therefore, some works [3,5] engage in incorporating semantic attributes with low-level features in image retrieval.

On the other hand, as image databases get large, a good organization is needed to aid better searching and browsing. Big image collections are often grouped into semantic classes, which may be further organized in a hierarchy. These semantic classes may be constructed manually, as did in most of the image search engines, or automatically using semantic attributes. Some efforts [1,14,15] attempted to automatically classify images into semantic classes, such as indoor, outdoor, man-made and natural, etc. WebSeek [13] performs a semi-automatic classification of images on the Web into a hierarchy of subjects by using associated text and filename cues.

A straightforward way to combine image classification with CBIR is to use classes as a filter to restrict the search range, and to reduce both searching time and potential retrieval errors. For example, when performing image search in WebSeek, the user first selects a category through the hierarchy of subjects, browses and selects an image as the query example. Then, the searching process is constrained in that category.

However, this method is not effective in some cases. In some image databases, the number of image categories is large. Although they can be organized in a hierarchy, it is not easy for a common user to find the intended category. When the user's need does not have apparent relations with these categories, it is difficult for the user to make a choice. So users tend to perform retrievals at a high level in the hierarchy. Due to the subjectivity and diversity of human perception of image content, a picture can mean different things to different people, and even mean different things to the same person at different time. Thus, certain classifications cannot satisfy all the different users and uses of images. Therefore, using image classes as a filter for image retrieval has a very limited success.

In this paper, we present an image retrieval scheme that integrates semantic classification with low-level features to improve the performance of content-based image retrieval systems. Class membership of an image is used as the semantic feature and is represented as a Boolean vector. Such class membership is modified through a relevance feedback process to better reflect user's perception of image content. This scheme may be

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considered a practical application of the hidden attribute approach described by Cox et al [5]. Experimental result shows the effectiveness of the proposed method.

The rest of the paper is organized as follows. The proposed image retrieval scheme is presented in Section 2. In section 3, we described the system implementation. The preliminary experimental results are presented in Section 4. Finally, we give discussion and concluding remarks in Section 5.

2. Image Retrieval Using Classification

Image classification can be regarded as treating a group of images as equivalent. Images within classes are more similar than those across classes in some way. If one or more images in a class are relevant to the user's query, other images in the same class are expected to be relevant in some extent as well.

In order to fully utilize the classification information existing in the image data, image classes are treated as pseudo keywords such that many text retrieval techniques may be deployed [2]. For the sake of simplicity, a Boolean vector is formed for every image with each element representing a class. If the image belongs to a particular class, the corresponding element in the vector is set to 1; it is set to 0 otherwise. This vector can be used as a semantic feature in addition to low-level features in image retrieval.

There are three main parts in the proposed scheme: feature representation, integration in retrieval, and relevance feedback. We only present the details related to the semantic feature in this section.

2.1 Feature Representation

Let t_{ij} denote the membership of image i to class j, i.e., $t_{ij}=1$ if $i \in j$, $t_{ij}=0$ if $i \notin j$. The semantic feature of image i is defined as below:

$$V_{ih} = (t_{i1}, t_{i2}, \cdots, t_{im})^T \quad (1)$$

where m is the total number of classes. In case that there is a class hierarchy, all high-level classes are also included.

As in text information retrieval, term weighting may be introduced in the feature representation, such as TF*IDF. Then, we get another definition:

$$V_{ih}' = (t_{i1} * \log(\frac{N}{n_1}), t_{i2} * \log(\frac{N}{n_2}), \cdots, t_{im} * \log(\frac{N}{n_m}))^T \quad (2)$$

where N is the total number of images in the database, n_j is the number of images in class j. The new definition implies that classes with more images are less useful in retrieval.

2.2 Retrieval with Classification

In our scheme, the visual similarity and the semantic similarity between two images are calculated separately. The former is based on the low-level features, while the latter is based on the class membership. The overall similarity is the weighted sum of these two similarities, as defined in Equation (3):

$$S(q,i) = \omega * S_h(q,i) + (1-\omega) * S_l(q,i) \quad (3)$$

where q is the query and i is an image, S_h and S_l are their semantic and visual similarities, respectively. ω is the semantic weight, $0 \le \omega \le 1$.

The semantic similarity between the query and an image is calculated by the cosine of the angle between these two vectors:

$$S_h(v_1, v_2) = \frac{v_1 \bullet v_2}{\|v_1\| \cdot \|v_2\|} \quad (4)$$

2.3 Relevance Feedback

Initially, the semantic feature of the query is not available if the query example is new to the system, thus all elements in the vector are set to 0. On the other hand, if the example is from the database, its semantic feature vector may be used directly. However, it is not necessarily consistent with the user's intention. Therefore, relevance feedback is deployed to iteratively refine the semantic feature vector of the query in order to better reflect the user's intention.

There are many feedback techniques proposed in the text retrieval field. Here, we simply adopt the Rochio formula [2] to refine the semantic feature of the query:

$$Q' = \alpha Q + \beta \frac{\sum Q^+}{n^+} - \gamma \frac{\sum Q^-}{n^-} \quad (5)$$

where Q is the original query, Q^+ is the set of positive examples, n^+ is the number of positive examples, while Q^- and n^- are that of negative examples respectively, Q' is the updated query. In our experiments, $\alpha = 0$, $\beta = 1$, $\gamma = 0.5$.

3. System Implementation

We integrated the proposed scheme into a prototype CBIR system, which adopts the query by example (QBE) mode. There are six low level visual features used in the system: 256D color histogram in HSV space, first and second color moments in Lab space, 64D color coherence vector in LUV space, MRSAR texture feature, Tamura coarseness feature, and Tamura directionality. Their effectiveness has been proven in previous works [7,8].

For relevance feedback on the low level features, a simpler version of the general relevance feedback

framework proposed by Rui [11] was deployed. During the retrieval, the user only marks the retrieved images as relevant or irrelevant.

A weighted Euclidean distance is adopted instead of a generalized one. In order to combine with the similarity based on the semantic features, the distance is further converted to similarity, which monotonously decreases as the distance increases. The similarity is normalized to the interval [0,1].

When classification is used in retrieval, the semantic feature vector is defined by Equation (1).

4. Experimental Results

In this section, we will compare the performance of the system adopting our scheme with that only based on low-level features. And further, to investigate the application of our scheme with different classification, we designed two experiments. In the first experiment, the classification in the Corel data set is used, which was constructed manually by domain specialists. In the second experiment, the classification was made automatically using feature-based classification algorithms.

4.1. Experimental Design

The image database we used consists of more than 10,000 images from the Corel dataset. It is a large and heterogeneous image set. Domain professionals classified all images in the dataset into semantic categories, each of which contains 100 images. Originally, there are 101 categories in the database. We merged some categories that share the same semantic meaning, such that each merged category depicts a distinct semantic topic. For example, category *Castles* in Disk07 and category *Castle II* in Disk08 are merged into one category, *Castle*. After that, we got a total of 79 categories.

We designed an automatic feedback scheme in the experiments. First, some query images are selected, and the ground truth is determined for each query image. These two steps will be described in section 4.2 and 4.3. After that, a reproducible relevance feedback for every query can be simulated based on the ground truth and the retrieval results of the system. At each iteration of the feedback, the system scans the first 30 retrieved images, and labels 3 irrelevant examples and at most 3 relevant examples, not including previous labeled images. Then the system use these labeled examples and that from previous iterations to conduct relevance feedback.

Precision-scope curve is used to evaluate the image retrieval performance. Scope specifies the number of images returned to the user. Precision is defined as the percentage of retrieved images that are actually relevant. Results with and without classification information are referred to as "Class" and "Orig" respectively.

4.2. Result Using Manual Classification

In this experiment, all 79 categories in the database are used as classification information. Thus, a 79-dimensional vector is formed for each image.

We selected 200 image examples as the query set. Most images are from the database, but 4 images are not. Then we collected judgments from 7 subjects as the ground truth. For every query example, each subject was required to scan the whole image set, and label all relevant ones based on his/her subjective judgment. In fact, the subjects selected different and varying numbers of relevant images for each query. Therefore, the retrieval precision was calculated for each subject separately and averaged finally, as if there were 1,400 different queries.

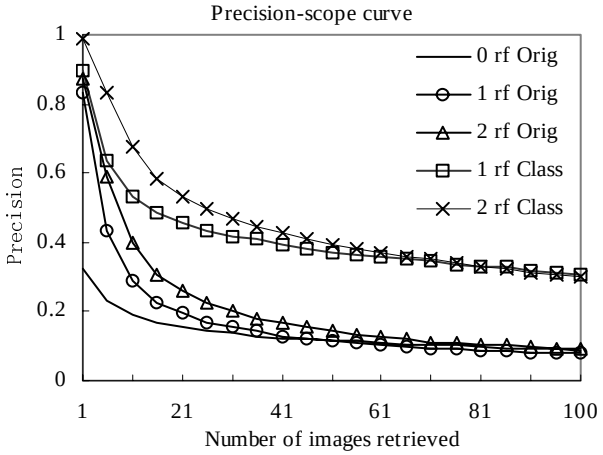


Figure 1. Average precision for the first two iterations.

The result of this experiment is presented in Figure 1, where "rf" stands for relevance feedback, and the digit before it stands for the number of iterations in feedback. The semantic weight is set to 0.2 in case that classification is used. The result shows that the proposed method can improve the precision in this experiment. P(20) of "Orig" without relevance feedback is 0.16. After the second iteration of relevance feedback, P(20) of "Orig" is 0.27, while that of "Class" is 0.54.

After investigating the data in detail, we also got some valuable observation. If most of the ground truth images of a query distribute in 1 or 2 categories, our method tends to make a greater improvement in precision. However, if the distribution of the ground truth images is dispersive, our method tends to make a smaller improvement, or even make a little drop. In other word, if the user's intention is consistent with the image classification, our method will likely make improvement.

Because the classification in Corel data set was

performed by domain professionals, it might match the perception of a common user very well. So much improvement is observed in the experiment when the classification information is integrated in the image retrieval.

4.3. Result Using Automatic Classification

In this experiment, we tested our method when using automatic image classification. Two SVM classifiers were trained to classify the images into semantic classes. Images were first classified to indoor or outdoor, and outdoor images were further classified to man-made or natural. This method achieved an accuracy of 91.6% for indoor vs. outdoor classification, while 93% for man-made vs. natural classification. It is comparable to other classification algorithms [14,15].

In this experiment, we chose 30 out of 79 categories, totally 4,400 images. The criterion of selecting these 30 categories is mainly to guarantee the accuracy of image classification, and exclude those images unrelated to the classification, such as images of "flag", or "marble".

The classification made by domain professionals was not used in composing the semantic feature, but as the ground truth. That is, a retrieved image was considered a match if it belongs to the same category of the query image. The automatic classification was used in forming a 4-dimensional semantic feature vector for each image. The semantic weight is set to 0.1. We tested 220 sample images chosen randomly from these 30 categories. To evaluate the performance, we averaged the precision over all queries. P(20) of "Orig" without relevance feedback is 0.28. After 2 iteration of relevance feedback, P(20) of "Orig" is 0.48, while P(20) of "Class" is 0.52. The improvement is small.

After analyzing the data, we found the observation in the previous experiment still holds. Since this kind of classification is very coarse and inconsistent with the user's intention in most cases, the result is reasonable.

5. Discussions and Conclusions

In this paper, we have proposed a scheme to effectively integrate the semantic classification with low-level features for content-based image retrieval. Although any kind of semantic classification can be used, the effect of our method does depend on the nature of the classification. If the classification information conforms the user's intention very well, the improvement in retrieval accuracy may be observed. Experimental results on a large heterogeneous image collection with manual classification show that the proposed method is effective.

We may further refine this method in the future. In Section 2, a fixed weight is used to combine the semantic feature and low-level features. Dynamic adjustment of the weight may make the approach more flexible. Furthermore, we may take account of the relationship between classes in image retrieval. In practice, this method may be used together with the traditional methods that use image classes as a filter.

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